

ECON 351 · Chapter 4 Practice Set

12 questions total

Set 1: Consumer Problems (2)

Set 2: Labor-Leisure (2)

Set 3: Intertemporal Consumption (2)

Set 4: Elasticity / Types of Goods (3)

Set 5: Budget Line (1)

Set 6: Assumptions / Preferences (2)

Consumer Problems

2 questions

Question 1 of 2 (Practice Q1)

Consumer Problem #1 · HW1 Ex 3

Exercise 3 - Optimal Consumption

3.1 Manuel has the following utility $U(q_1, q_2) = 3q_1^{0.6}q_2^{0.4}$. What values of q_1 and q_2 maximize his utility?

3.2 Now suppose prices are $p_1 = 5$ and $p_2 = 10$ and that Manuel has \$100. Write Manuel's budget constraint and graph it with q_1 in the x-axis. What is the slope? And the intercept? What is the maximum amount of q_1 he can purchase? And the maximum amount of q_2 he can purchase?

3.3 Next, we will find out the best use Manuel could make of his money, given his preferences. Based on Manuel's preferences, let's figure out the ratio between q_1 and q_2 . What is the ratio of the two marginal utilities (MRS)? (remember that MU_1 is in the numerator)

3.4 Now we have the ratio of how Manuel trades both goods. We will equalize it to the ratio of both prices. That is, how Manuel trades both goods needs to equal how the market trades them. In the optimal bundle, how do the marginal utilities relate to market prices?

3.5 Finally, use the previous relationship to substitute in the budget constraint and find the optimal quantities of q_1 and q_2 , which we denote by q_1^* and q_2^* .

Question 2 of 2 (Practice Q2)

Consumer Problem #2 · Sample Q26

Question 26:

There are two goods, apples and bananas. The price of apples is $P_A = \$2$, and the price of bananas is $P_B = \$3$. A consumer has \$120 to spend, and his utility function is $U(A, B) = 2A^2B^3$.

a) Write the expression that represents the consumer's budget constraint. Graph the budget constraint, and label it "BC" (apples should be on the horizontal axis). What is the slope of the budget constraint?

b) Find the consumption basket that maximizes the consumer's utility. Plot this point on the graph, and label it as "Point X". Draw the indifference curve that goes through Point X, and label it "IC" (the indifference curve can be an approximation, it does not have to be exact). What is the slope of the indifference curve at Point X?

Question 27:

Adam derives utility from consuming apples and bananas. His marginal utility from consuming apples is

$$MU_A = 5A^4B^3,$$

while his marginal utility from consuming bananas is

$$MU_B = 3A^5B^2,$$

where A is the quantity of apples and B is the quantity of bananas. We know that the price of apples is $P_A = 6$, the price of bananas is $P_B = 2$ and Adam's income is $I=240$. Given this information, when Adam optimally chooses his consumption, he buys how many units of apples?

(A) A=65

(B) A=45

(C) A=25

(D) A=35

(E) A=55

Question 28:

Monica only consumes goods A and B. Suppose that her marginal utility from consuming good A is equal to $1/Q_A$, and her marginal utility from consuming good B is $1/Q_B$. If the price of A is \$0.50, the price of B is \$4.00, and the Monica's income is \$120.00, how much of good A will she purchase?

(A) 0

(B) 12

(C) 24

(D) 48

(E) 120

Labor-Leisure

2 questions

Set 2: Labor-Leisure

Question 1 of 2 (Practice Q3)

Labor-Leisure #1 · Course-style

Maria: 16 hrs/day, $w=\$20$, $V=\$80$, $U=c^{0.5} l^{0.5}$. Find I^* , L^* , c^* .
(Course-style)

Set 2: Labor-Leisure

Question 2 of 2 (Practice Q4)

Labor-Leisure #2 · Course-style

Same setup, $w=\$30$. Find I^* , L^* , c^* . More or less work?

Intertemporal Consumption

2 questions

Question 1 of 2 (Practice Q5)

Intertemporal #1 · Sample Q41

Question 38:

A consulting company researched the market for apples in California, and concluded the following: apples and bananas are complements, and apples are an inferior good.

Define the following:

Q_A : quantity of apples demanded

P_A : price of Apples

P_B : price of bananas

I : income

From the options below, mark the only demand function that can possibly represent the demand for apples in California.

(A) $Q_A = 100 - 3P_A + 3I - 5P_B$

(B) $Q_A = 200 - P_A + 2I + 4P_B$

(C) $Q_A = 110 - 2P_A - 4I + 2P_B$

(D) $Q_A = 250 - 4P_A - 5I - 3P_B$

(E) $Q_A = 150 + P_A + I + P_B$

Question 39:

The demand for apples (Q_A) is given by the function

$$Q_A = 784 - 2P_A - 3I - 4P_O$$

where P_A is the price of apples, I is the income level, and P_O is the price of oranges.

Suppose $P_A = 2$, $I = 100$, and $P_O = 20$.

a) Compute the price elasticity of demand for apples.

b) Compute the income elasticity of demand for apples. Are apples a normal good or inferior good?

c) Compute the cross-price elasticity of demand for apples. Are apples and oranges substitutes or complements?

Question 40:

The income elasticity of demand for good X is -2. Due to a financial crisis, the income of consumers declined by 2%. How did the financial crisis affect the quantity demanded for X?

(A) The quantity demanded increased 2%

(B) The quantity demanded decreased 2%

(C) The quantity demanded increased 4%

(D) The quantity demanded decreased 4%

(E) The quantity demanded did not change

Question 41:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 50%, that is, $r = 0.5$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ? Is this consumer saving or borrowing? How much?

Question 42:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 5%, that is, $r = 0.05$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ?
Is this consumer saving or borrowing? How much?

Question 2 of 2 (Practice Q6)

Intertemporal #2 · Mock Q24-25

Question 24:

[Show your work]

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 60$ in the first period and $I_2 = 125$ in the second period. The consumer can borrow or save money at the interest rate of 25%, that is, $r=0.25$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

The consumer's inter-temporal budget constraint is

- (A) $q_1 + 0.8q_2 = 160$
- (B) $q_1 + q_2 = 185$
- (C) $q_1 + 1.25q_2 = 216.25$
- (D) $q_1 + 0.75q_2 = 160$
- (E) $q_1 + 0.25q_2 = 200$

Question 25:

[Show your work]

Consider all the information from Question 24. In the first period, the consumer will optimally purchase

- (A) $q_1 = 0$
- (B) $q_1 = 100$
- (C) $q_1 = 60$
- (D) $q_1 = 80$
- (E) $q_1 = 120$

Elasticity / Types of Goods

3 questions

Question 1 of 3 (Practice Q7)

Elasticity #1 · HW1 Ex 4

Exercise 4 - Price Elasticity

Here, we will review the concept of elasticity and go through some examples. Price elasticity of demand is how much quantity demanded changes when the price goes up by 1%. Reminder $E_p = \frac{P}{Q} \frac{\partial Q}{\partial P}$

Take the following demand function $q = 100 - 5p$.

4.1 Calculate the quantity demanded when $p = 5$ and when $p = 10$. What happens to the quantity demanded when the price increases? Given your answer, do you expect the sign of price elasticity to be positive or negative?

4.2 Calculate the price elasticity when $p = 5$. Interpret the elasticity.

4.3 Calculate the quantity demanded when $p = 5.05$.

4.4 Note that $p = 5.05$ corresponds to a 1% price increase with respect to $p = 5$. Compare this % increase with the % change in the quantity demanded. That is, calculate $\Delta Q / Q(p = 5) * 100$. How does this compare to the price elasticity you calculated before?

4.5 Redo points 4.2-4.4 but now calculate the elasticity at $p = 10$ and compare the % change in quantity demanded when $p = 10.1$. How does the % change in quantity compared to the elasticity at $p = 10$?

4.6 EXTRA In the previous questions, you have calculated the elasticity using the formula and also by "brute force". In these cases, the values coincide, but in other cases, there would be a small gap between both methods. Why is that the case, and why are they the same here? (Hard question. Tip: it has to do with the basic concept of a derivative and the curvature of the demand function.)

Question 2 of 3 (Practice Q8)

Elasticity #2 · Sample Q39

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(A) $Q_A = 100 - 3P_A + 3I - 5P_B$

(B) $Q_A = 200 - P_A + 2I + 4P_B$

(C) $Q_A = 110 - 2P_A - 4I + 2P_B$

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Suppose $P_A = 2$, $I = 100$, and $P_O = 20$.

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Question 3 of 3 (Practice Q9)

Elasticity #3 · Sample Q38

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Question 42:

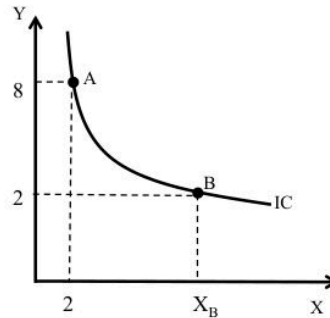
Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 5%, that is, $r = 0.05$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

Budget Line

1 question

Question 1 of 1 (Practice Q10)

Budget Line · Sample Q10

What do we know about X_B ?

- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

Question 10:

Assume that apples are measured on the horizontal axis and oranges on the vertical axis. If the price of apples increases from \$3 to \$6, and the price of oranges increases from \$2 to \$4, then the budget line will:

- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).
- (E) become steeper or flatter depending on the relationship between prices and income.

Question 9:

Angelique's preferences over goods X and Y can be represented by the following utility function: $u(X, Y) = X^2Y^3$. The graph below represents two baskets, A and B, on the same indifference curve of this consumer.

Assumptions / Preferences

2 questions

Question 1 of 2 (Practice Q11)

Preferences #1 · HW1 Ex 2.2

Exercise 2 - Preferences' Assumptions

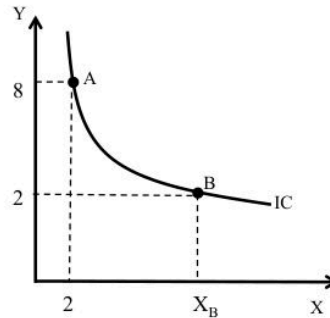
2.1 Jackie was at the movie theater last night. The two movies available were "Knives out" and "Top Gun Maverick." She didn't care which one to see. Does this imply a violation of any assumption on preferences? If so, which one?

2.2 Again, Jackie goes to the movies. The two movies available are "Avatar" and "The Avengers," and she says she cannot decide which one to see. Does this imply a violation of any assumption on preferences? If so, which one?

2.3 On a different day, there were three movies available: "Parasite," "Indiana Jones," and "Jurassic Park." Jackie said she rather watch Parasite over Indiana Jones and Indiana Jones over Jurassic Park. But she would also prefer watching Jurassic Park over Parasite. Does this imply a violation of any assumption on preferences (beyond displaying poor taste)? If so, which one?

Question 2 of 2 (Practice Q12)

Preferences #2 · Sample Q8

What do we know about X_B ?

- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

Question 10:

Assume that apples are measured on the horizontal axis and oranges on the vertical axis. If the price of apples increases from \$3 to \$6, and the price of oranges increases from \$2 to \$4, then the budget line will:

- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).
- (E) become steeper or flatter depending on the relationship between prices and income.

Question 9:

Angelique's preferences over goods X and Y can be represented by the following utility function: $u(X, Y) = X^2Y^3$. The graph below represents two baskets, A and B, on the same indifference curve of this consumer.